Network Replicability & Generalizability: Exploring the Effects of Sampling Variability, Scale Variability, and Node Aggregation

Supplementary Materials

Network Comparison Results

Table S1

Differences in Adjacency Matrices: Descriptives [a.]

	Absent in both		Present in both		Different in each	
	***n* = 84**	***n* = 212**	***n* = 84**	***n* = 212**	***n* = 84**	***n* = 212**
Sampling variability						
1-item	7.66	6.12	2.22	3.98	5.12	4.90
2-item	6.26	4.84	3.32	5.26	5.42	4.90
3-item	5.66	4.76	3.64	6.08	5.70	4.16
5-item	5.76	4.50	4.24	6.94	5.00	3.56
8-item	5.64	4.20	4.84	7.06	4.52	3.74
Scale variability						
	Absent in both		**Present in both**		**Different in each**	
	***n* = 84**	***n* = 212**	***n* = 84**	***n* = 212**	***n* = 84**	***n* = 212**
1-item	7.40	5.38	1.58	3.56	6.02	6.06
2-item	6.20	4.48	2.98	4.86	5.82	5.66
3-item	5.66	4.42	3.58	5.48	5.76	5.10
5-item	5.36	4.34	4.16	6.68	5.48	3.98
8-item	5.30	4.26	4.80	6.78	4.90	3.96
Sampling and scale variability						
	Absent in both		**Present in both**		**Different in each**	
	***n* = 84**	***n* = 212**	***n* = 84**	***n* = 212**	***n* = 84**	***n* = 212**
1-item	7.24	5.64	1.80	3.18	5.96	6.18
2-item	5.92	4.22	3.02	4.76	6.06	6.02
3-item	5.56	4.28	3.42	5.82	6.02	4.90
5-item	5.40	4.20	4.24	6.34	5.36	4.46
8-item	5.24	4.50	4.50	6.50	5.26	4.00

a. Values correspond to Figure 1 in manuscript.

Table S2

Average Global Strength: Descriptives [a]

	NEO scale				IPIP scale			
	***n* = 84**	***n* = 212**	***n* = 339**	***N* = 424**	***n* = 84**	***n* = 212**	***n* = 339**	***N* = 424**
1-item	1.32	1.50	1.50	1.57	1.44	1.59	1.61	1.57
2-item	1.84	1.96	1.92	1.97	1.87	1.94	2.00	2.01
3-item	2.10	2.10	2.17	2.15	2.05	2.12	2.15	2.18
5-item	2.25	2.29	2.31	2.30	2.27	2.29	2.31	2.31
8-item	2.38	2.39	2.39	2.41	2.41	2.39	2.38	2.38

a. Values correspond to Figure 2 in manuscript.

Table S3

Differences in Average Global Strength: Descriptives [a]

	Sampling variability		Scale variability		Sampling and scale variability	
	***n* = 84**	***n* = 212**	***n* = 84**	***n* = 212**	***n* = 84**	***n* = 212**
1-item	0.38	0.19	0.37	0.27	0.39	0.28
2-item	0.25	0.14	0.24	0.18	0.28	0.19
3-item	0.30	0.11	0.18	0.14	0.28	0.15
5-item	0.21	0.11	0.18	0.10	0.20	0.12
8-item	0.18	0.07	0.20	0.07	0.23	0.08

a. Values correspond to Figure 3 in manuscript.

Table S4

Correlations Between Edge Weights: Descriptives [a]

	Sampling variability		Scale variability		Sampling and scale variability	
	***n* = 84**	***n* = 212**	***n* = 84**	***n* = 212**	***n* = 84**	***n* = 212**
1-item	0.30	0.50	0.09	0.25	0.17	0.26
2-item	0.38	0.60	0.24	0.31	0.20	0.33
3-item	0.43	0.66	0.30	0.47	0.33	0.46
5-item	0.50	0.70	0.41	0.60	0.41	0.52
8-item	0.52	0.71	0.57	0.66	0.49	0.61

a. Values correspond to Figure 4 in manuscript.

Table S5

Correlations Between Centrality Scores (Expected Influence): Descriptives [a]

	Sampling variability		**Scale variability**		**Sampling and scale variability**	
	***n* = 84**	***n* = 212**	***n* = 84**	***n* = 212**	***n* = 84**	***n* = 212**
1-item	0.42	0.53	0.17	0.47	0.23	0.31
2-item	0.51	0.70	0.46	0.57	0.39	0.53
3-item	0.63	0.72	0.55	0.70	0.39	0.62
5-item	0.64	0.81	0.64	0.81	0.65	0.78
8-item	0.69	0.85	0.71	0.89	0.69	0.86

a. Values correspond to Figure 5 in manuscript.

Centrality: Strength

Table S6

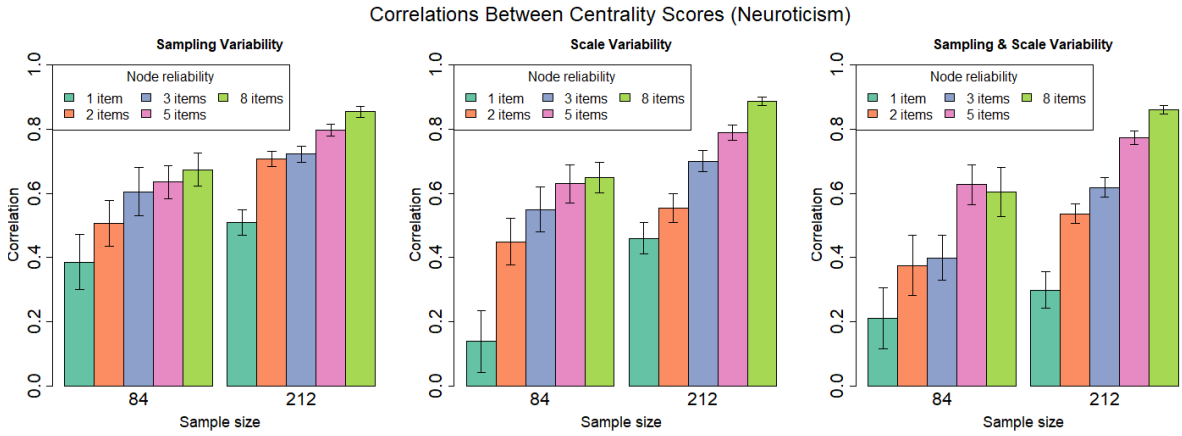
Correlations Between Centrality Scores (Strength): Descriptives [a]

	Sampling variability		Scale variability		Sampling and scale variability	
	***n* = 84**	***n* = 212**	***n* = 84**	***n* = 212**	***n* = 84**	***n* = 212**
1-item	0.39	0.51	0.14	0.46	0.21	0.30
2-item	0.51	0.71	0.45	0.55	0.38	0.54
3-item	0.61	0.72	0.55	0.70	0.40	0.62
5-item	0.64	0.80	0.63	0.79	0.63	0.77
8-item	0.67	0.85	0.65	0.89	0.60	0.86

a. Values correspond to supplementary Figure S1 (below).

Figure S1

Correlations Between Centrality Scores (Strength)



Note. Effects of sample size and node aggregation on rank correlations between centrality (i.e. Strength) scores, across resampling conditions: sampling variability (left), scale variability (middle), sampling and scale variability (right). Error bars represent 95% confidence intervals around the mean.

Table S7

Whole-Sample Networks: Centrality (Strength scores)

Network characteristic		NEO 8-items	IPIP 10-items
CS-coefficient for Strength [a]		0.75	0.75
Centrality ranks (Strength raw score)			
	1st	DEP (1.17)	ANX (1.15)
	2nd	VUL (0.91)	DEP (1.12)
	3rd	ANX (0.89)	VUL (0.85)
	4th	ANG (0.70)	ANG (0.65)
	5th	SELF (0.68)	SELF (0.58)
	6th	IMP (0.48)	IMP (0.45)

Note. All values were computed on the full sample of N = 424. ANX = anxiety, ANG = anger, DEP = depression, SELF = self-consciousness, IMP = impulsiveness, and VUL = vulnerability.
a. *CS*-coefficients are an index of centrality stability, and represent the maximum proportion of sampled cases that can be dropped in order to retain a correlation of 0.7 (with 95% probability) between original sample and subsample centrality estimates (Epskamp et al., 2018).

Note. Strength scores and Expected Influence scores are identical because all edges are positive in the whole-sample networks.

Table S8

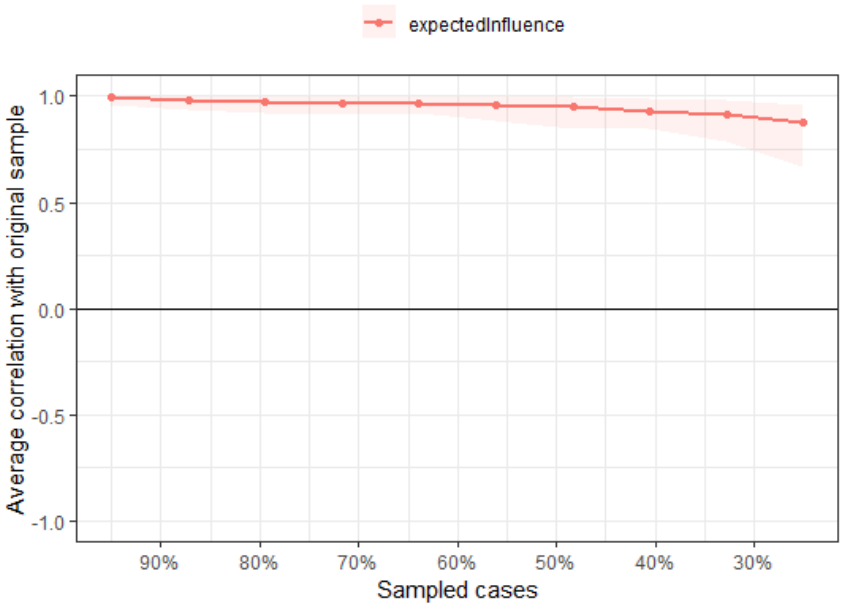
Scale Variability Condition: Descriptive Overview of Network Characteristics (Strength scores)

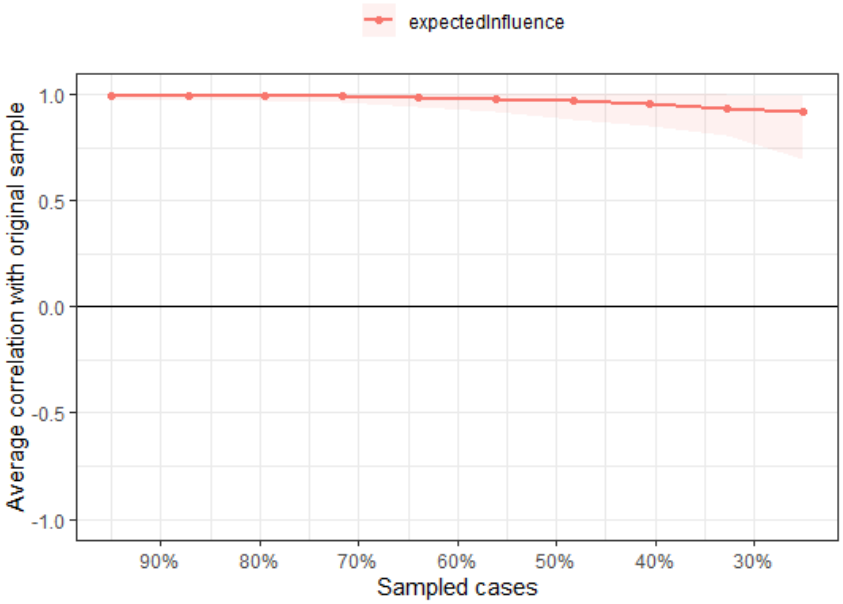
Network characteristic Mean (SD)	1-item		3-items		8-items	
	NEO	IPIP	NEO	IPIP	NEO	IPIP
Avg. correlation between centrality scores (Strength)	r_s = .46 (.36)		r_s = .70 (.25)		r_s = .89 (.10)	
Centrality ranks: Strength (frequencies, out of 50, listed in parentheses)						
Most central node (i.e. ranked 1st most often)	DEP (29)	DEP (23)	DEP (36)	DEP (32)	DEP (49)	DEP (31)
Second most central node (i.e. ranked 1st second most often)	ANX (10)	ANX (13)	VUL (8)	ANX (13)	VUL (1)	ANX (17)
Third most central node (i.e. ranked 1st third most often)	SELF (6)	VUL (9)	ANX (5)	VUL (5)	n/a	VUL (2)

Note. All values were computed on a sample of n = 212. ANX = anxiety, ANG = anger, DEP = depression, SELF = self-consciousness, IMP = impulsiveness, and VUL = vulnerability. r_s denotes Spearman-rank correlation coefficients.

Figure S2

Whole-Sample Networks: Centrality Stability (Expected Influence)

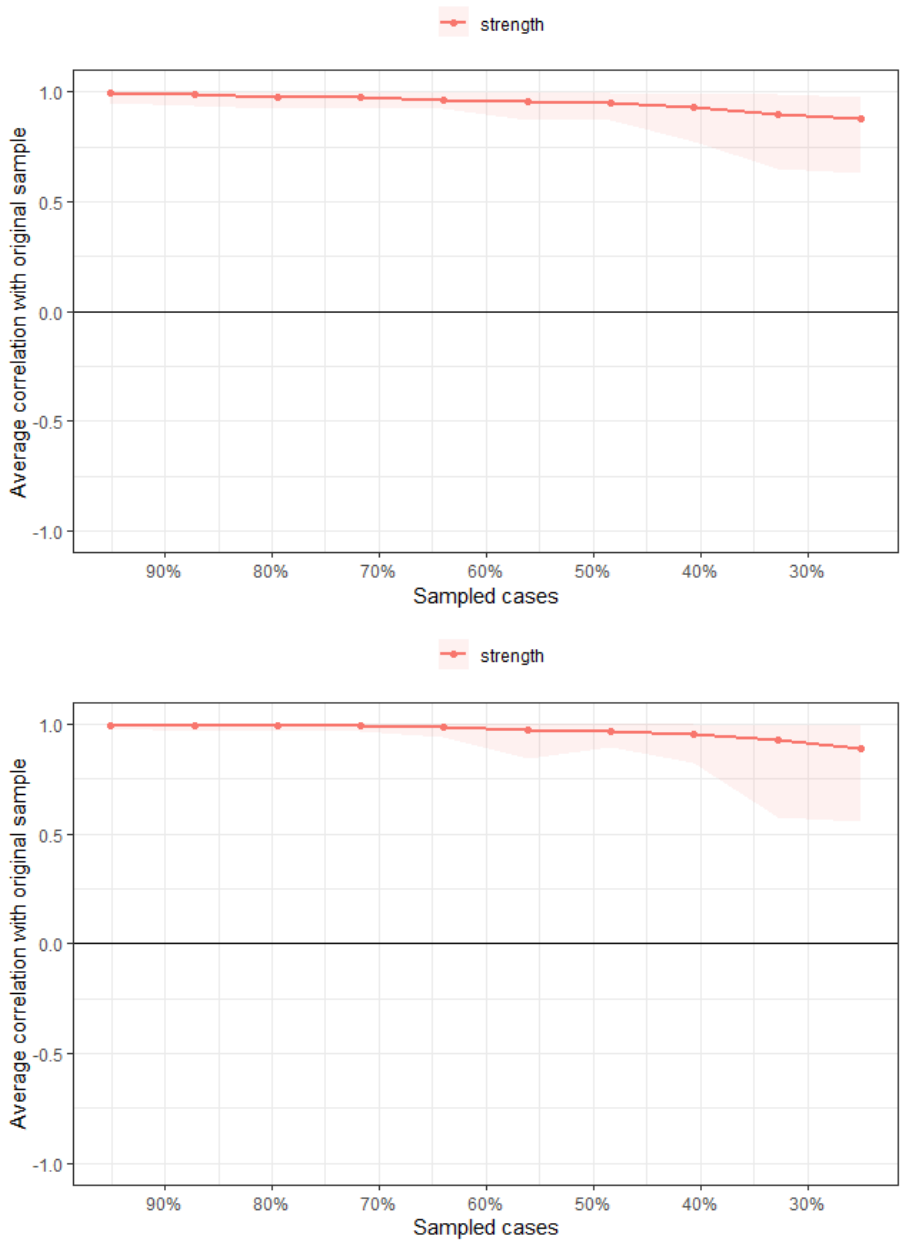




Note. Stability in Expected Influence scores are illustrated for the NEO-PI-R (top plot) and IPIP (bottom plot) whole-sample networks. X-axis represents proportions of sampled cases, and y-axis correlation coefficient values. Average correlations between centrality scores, estimated on the original sample and on the subsamples, are indicated by the solid line; shaded area indicates the range from the 2.5th to the 97.5th quantiles.

Figure S3

Whole-Sample Networks: Centrality Stability (Strength)



Note. Stability in Strength scores are illustrated for the NEO-PI-R (top plot) and IPIP (bottom plot) whole-sample networks. X-axis represents proportions of sampled cases, and y-axis correlation coefficient values. Average correlations between centrality scores, estimated on the original sample and on the subsamples, are indicated by the solid line; shaded area indicates the range from the 2.5th to the 97.5th quantiles.

Exploratory Factor Analysis (EFA)

Table S6

EFA 6-Factor Solution (NEO): Pattern Matrix (standardized loadings)

Items (NEO)	Factor					
	1	2	3	4	5	6
N1_1	-0.03	0.10	0.04	0.01	**0.65**	0.01
N1_2	0.03	-0.09	0.00	0.38	0.35	0.01
N1_3	0.03	0.02	0.08	0.14	**0.55**	0.26
N1_4	0.16	0.19	-0.01	0.11	0.23	0.30
N1_5	0.00	0.02	0.07	-0.05	**0.57**	0.17
N1_6	0.15	0.13	0.04	0.06	**0.54**	-0.01
N1_7	0.01	-0.13	0.03	0.21	0.35	0.25
N1_8	0.08	0.12	0.08	0.11	0.26	0.05
N2_1	0.33	**0.52**	-0.12	-0.04	0.10	-0.01
N2_2	-0.07	**0.70**	0.06	0.03	-0.02	0.08
N2_3	-0.07	**0.74**	0.04	-0.03	-0.02	-0.04
N2_4	0.02	**0.54**	-0.02	0.11	0.02	0.03
N2_5	0.19	**0.42**	0.02	-0.12	0.13	0.00
N2_6	-0.09	**0.63**	0.01	0.08	0.06	-0.01
N2_7	0.23	0.34	0.11	-0.07	0.11	-0.06
N2_8	0.17	**0.40**	0.01	0.09	0.12	0.00
N3_1	0.10	0.09	-0.06	-0.01	0.22	**0.59**
N3_2	**0.55**	0.06	0.07	0.01	0.02	0.29
N3_3	0.08	0.10	0.04	0.00	0.22	**0.63**
N3_4	**0.43**	-0.01	0.16	0.02	-0.10	0.13
N3_5	**0.51**	-0.02	0.01	0.04	0.10	-0.08
N3_6	**0.51**	-0.03	0.08	0.13	0.06	0.22
N3_7	0.33	0.14	0.04	0.02	0.07	**0.43**
N3_8	**0.50**	0.10	0.03	0.19	-0.06	0.17
N4_1	**0.49**	-0.04	-0.07	-0.01	0.27	-0.07
N4_2	0.30	-0.04	0.07	0.06	0.34	-0.08
N4_3	**0.55**	0.13	0.15	-0.03	-0.12	0.10
N4_4	0.20	0.06	-0.13	0.05	0.18	0.03
N4_5	**0.64**	-0.02	0.02	0.11	0.17	0.02
N4_6	0.28	0.00	0.01	0.27	0.07	0.05
N4_7	**0.60**	-0.08	0.02	0.01	0.10	-0.14
N4_8	0.19	-0.10	-0.03	0.07	0.19	-0.16
N5_1	-0.07	0.20	**0.57**	-0.12	0.07	0.09
N5_2	0.07	-0.06	**0.76**	0.07	0.05	-0.09
N5_3	-0.10	-0.11	**0.49**	0.05	0.04	0.20
N5_4	0.11	0.09	**0.52**	-0.08	0.12	-0.26

N5_5	-0.20	0.05	**0.42**	0.03	-0.11	0.20
N5_6	0.21	0.14	0.31	0.08	-0.03	-0.12
N5_7	0.31	0.11	0.37	0.04	-0.21	0.02
N5_8	-0.13	0.29	0.24	0.18	-0.09	0.11
N6_1	0.28	0.07	0.09	0.35	-0.03	0.21
N6_2	0.05	0.01	-0.01	**0.43**	0.06	0.29
N6_3	0.35	0.06	-0.02	0.17	0.02	0.28
N6_4	-0.04	0.07	0.01	**0.76**	0.01	-0.11
N6_5	0.21	-0.06	0.12	0.26	0.17	0.02
N6_6	-0.02	0.01	-0.02	**0.75**	0.04	-0.08
N6_7	0.11	-0.03	0.05	**0.59**	-0.07	0.12
N6_8	0.07	0.33	-0.01	0.25	0.00	0.33

Note. Six-factor solution for Neuroticism dimension (NEO questionnaire); *N* = 424 (whole sample), method = minres (minimum residual), rotation = oblimin. Loadings equal to or greater than 0.40 are in boldface type.

Table S7

EFA 6-Factor Solution (NEO): Factor Correlations

Factors	**1**	**2**	**3**	**4**	**5**	**6**
1	1.00	0.26	0.26	0.37	0.45	0.35
2		1.00	0.32	0.21	0.25	0.34
3			1.00	0.19	0.14	0.15
4				1.00	0.32	0.32
5					1.00	0.35
6						1.00

Table S8

EFA 6-Factor Solution (IPIP): Pattern Matrix (standardized loadings)

Items (IPIP)	**Factor**					
	1	**2**	**3**	**4**	**5**	**6**
N1_1	0.29	0.20	-0.05	0.10	0.04	0.25
N1_2	0.06	0.03	0.15	0.27	0.17	0.22
N1_3	0.15	0.16	0.02	0.26	0.09	0.20
N1_4	0.23	0.07	0.15	0.21	0.16	0.24
N1_5	0.32	0.14	-0.13	0.05	0.16	0.09
N1_6	0.34	-0.05	0.05	0.27	0.19	-0.01
N1_7	**0.46**	0.12	0.02	0.09	0.06	-0.04

N1_8	0.34	0.12	-0.03	0.39	-0.08	0.24
N1_9	0.29	0.03	0.14	0.22	0.03	0.32
N1_10	0.09	0.28	0.01	0.22	0.15	**0.43**
N2_1	0.02	**0.48**	0.07	0.07	0.01	0.13
N2_2	0.01	**0.78**	0.04	0.00	-0.04	-0.07
N2_3	-0.07	**0.69**	0.07	-0.02	0.02	-0.09
N2_4	-0.05	**0.76**	0.02	-0.02	0.14	-0.10
N2_5	-0.01	0.39	0.00	0.49	-0.10	0.06
N2_6	0.00	**0.59**	-0.05	0.07	0.09	0.12
N2_7	0.08	**0.63**	0.06	0.01	-0.03	0.08
N2_8	0.09	**0.66**	0.00	0.04	-0.01	0.10
N2_9	**0.51**	**0.40**	-0.02	-0.11	0.02	0.02
N2_10	0.19	**0.46**	0.07	0.20	-0.06	0.18
N3_1	**0.43**	0.29	0.12	0.01	-0.02	0.09
N3_2	**0.64**	0.12	0.13	-0.06	0.01	-0.09
N3_3	**0.51**	0.03	0.02	0.00	0.28	-0.26
N3_4	**0.49**	0.00	0.01	0.05	0.18	-0.03
N3_5	**0.68**	0.04	-0.06	0.06	-0.08	-0.14
N3_6	**0.52**	-0.02	0.06	0.07	0.14	0.09
N3_7	**0.55**	-0.01	0.16	0.12	0.14	-0.09
N3_8	**0.65**	0.06	0.11	-0.03	0.05	0.20
N3_9	**0.50**	-0.14	0.16	0.08	0.10	0.03
N3_10	**0.76**	0.03	0.05	-0.03	0.03	0.17
N4_1	0.21	-0.13	0.03	0.28	0.22	-0.19
N4_2	-0.03	0.08	0.18	0.07	0.38	-0.21
N4_3	0.13	0.07	0.00	-0.13	**0.59**	-0.08
N4_4	0.10	-0.05	-0.02	0.01	**0.50**	-0.16
N4_5	0.11	0.14	-0.03	-0.10	**0.43**	-0.03
N4_6	0.13	0.01	0.11	0.12	**0.42**	0.01
N4_7	0.13	-0.01	0.05	0.30	0.39	-0.21
N4_8	0.03	0.01	-0.02	0.08	**0.53**	0.22
N4_9	-0.05	0.08	0.01	0.00	**0.58**	0.18
N4_10	-0.05	0.09	0.02	0.14	**0.55**	0.07
N5_1	0.12	0.17	0.31	0.08	0.09	-0.19
N5_2	-0.05	0.06	**0.41**	0.06	-0.22	-0.12
N5_3	0.16	0.07	0.36	0.25	-0.08	-0.23
N5_4	-0.17	0.01	**0.45**	-0.08	0.05	0.15
N5_5	0.11	-0.03	**0.46**	0.12	-0.04	-0.08
N5_6	0.01	0.02	**0.64**	-0.04	0.10	0.05
N5_7	-0.03	0.08	**0.68**	-0.04	-0.05	0.04
N5_8	0.05	0.04	**0.44**	0.03	-0.25	-0.05
N5_9	0.07	-0.06	**0.55**	-0.04	0.13	0.02
N5_10	0.05	0.09	**0.48**	0.07	-0.04	-0.11
N6_1	-0.14	0.12	0.03	**0.60**	0.03	0.04
N6_2	0.26	0.10	-0.13	**0.44**	0.03	-0.18
N6_3	0.34	-0.02	-0.07	0.31	0.05	0.04

N6_4	-0.03	0.02	-.06	0.51	0.23	-0.19
N6_5	0.13	-0.08	0.16	0.26	0.25	-0.19
N6_6	0.17	0.01	0.10	**0.44**	0.14	0.02
N6_7	0.16	0.01	0.18	**0.40**	-0.03	0.05
N6_8	0.54	-0.01	-0.03	0.18	0.07	-0.18
N6_9	0.27	-0.03	0.05	**0.44**	0.07	0.03
N6_10	-0.06	0.02	0.06	**0.72**	0.00	0.07

Note. Six-factor solution for Neuroticism dimension (IPIP questionnaire); *N* = 424 (whole sample), method = minres (minimum residual), rotation = oblimin. Loadings equal to or greater than 0.40 are in boldface type.

Table S9

EFA 6-Factor Solution (IPIP): Factor Correlations

Factors	1	2	3	4	5	6
1	1.00	0.41	0.36	0.45	0.44	0.06
2		1.00	0.33	0.35	0.18	0.23
3			1.00	0.24	0.15	0.03
4				1.00	0.32	0.09
5					1.00	-0.05
6						1.00

Confirmatory Factor Analysis (CFA)

Table S10

CFA 6-Factor Solution: Fit Indices (for NEO & IPIP questionnaires)

Model	χ^2	df	*p*	CFI	TLI	AIC	BIC	RMSEA	SRMR
NEO	2625.43	1065	<.001	0.79	0.78	51157.02	51606.54	0.06	0.07
IPIP	3717.88	1695	<.001	0.80	0.79	67626.70	68173.41	0.05	0.07

Note. Criteria indicating good model fit CFI and TLI > .90, RMSEA and SRMR < .05 (Hu & Bentler, 1999; Kline, 2015)

Replication of Project using Agreeableness Data[1]

In the manuscript (Herrera-Bennett & Rhemtulla, 2021; https://psyarxiv.com/7vkm8/), we explored the effects of different sources of variability on consistency in network properties, across replications, specifically in the context of 6-node Neuroticism networks (constituent facets/nodes: Anxiety, Anger, Depression, Self-consciousness, Impulsiveness, Vulnerability). In order to verify whether observed trends held for different network constructs, we ran a replication of the current project on an additional personality dimension, namely 6-node Agreeableness networks (constituent facets/nodes: Trust, Straightforwardness, Altruism, Compliance, Modesty, Tender-Mindedness). See Appendix for all Agreeableness items.

First, we present the results of the resampling conditions, which explored the individual and joint effects of sampling variability and scale variability, at varying degrees of node aggregation. Namely, we look at: differences in adjacency matrices, average global strength, differences in average global strength, correlations between edge weights, and correlations between centrality scores. This is followed by a more focused look at the effects of scale variability (i.e. NEO scale vs. IPIP scale). Specifically, we compared network characteristics when Agreeableness networks were estimated on the whole-sample (using descriptive statistics, *Network Comparison Test* (*NCT*) results, and correlational analyses). Next, we characterized the variability in network properties, across replications, using a tallied frequency approach.

[1] A big thanks to Jiefei Li in helping to run and compile the Agreeableness replication.

Effects of Resampling Conditions

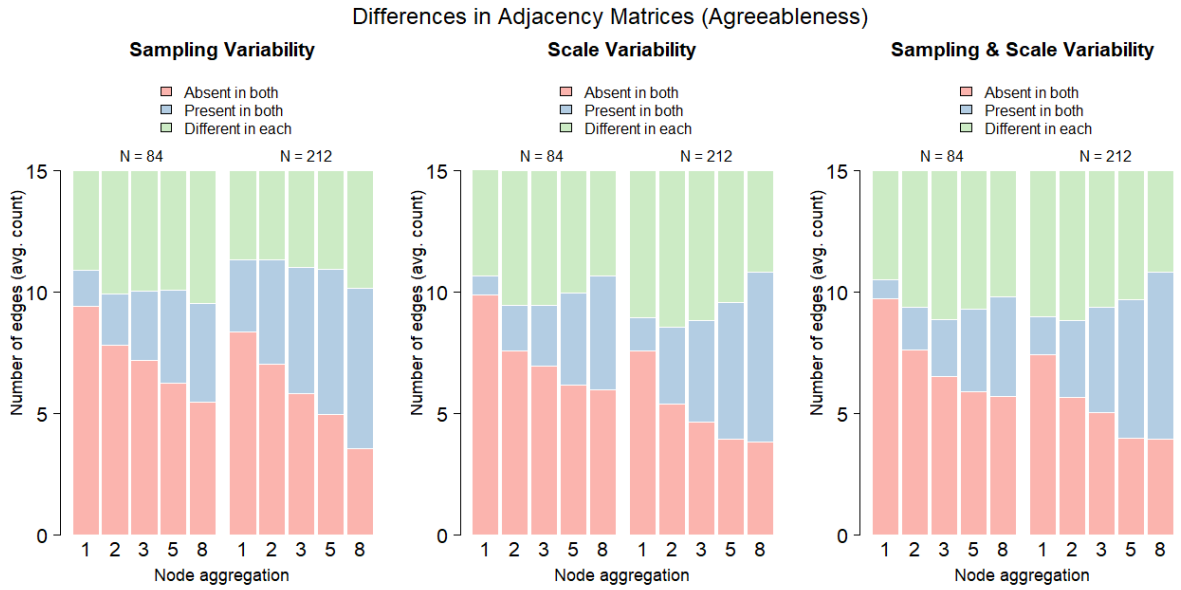
Overall, trends observed across resampling conditions for the Agreeableness data were consistent with those observed for the Neuroticism data. Use of multi-item indicators led to denser networks (Fig. A1), greater estimates of global strength (Fig. A2), and greater levels of consistency between edge weights (Fig. A4) and centrality scores (Fig. A5). With respect to resampling conditions, scale variability (or the combination of scale and sampling variability) produced less consistency in network properties than when only sampling variability was present. Interestingly, for the Agreeableness data, we observed that under more stable measurement conditions (i.e. sampling variability condition, n = 212, 8-item indicators), more pronounced differences in average global strength were observed as compared to lower levels of node aggregation (Fig. A3, left panel). This is inconsistent with the Neuroticism data which showed smaller discrepancies in global strength.

This can be explained by the fact that in the whole-sample Agreeableness networks, there was a greater difference in global strength scores (NEO network = 2.43; IPIP network = 2.60), whereas in the whole-sample Neuroticism networks, global strength scores were nearly identical (NEO network = 2.41; IPIP network = 2.40). As such, when networks were estimated on large samples and at high node aggregation (i.e., n = 212, 8-item), we would expect more representative reflections of the population networks (i.e., whole-sample networks which should exemplify any scale-specific differences). Because whole-sample Neuroticism networks had similar global strength scores, higher levels of node aggregation led to diminished discrepancies between these scores; in contrast, because whole-sample Agreeableness networks had more dissimilar global strength scores, higher levels of node aggregation highlighted these

discrepancies between scores. Although neither of the differences in global strength scores were deemed significant by the *NCT* invariance test (Agreeableness difference = 0.17, *p* = .177; Neuroticism difference = 0.01, p = .731), the observed pattern of results is still revealing: Conditions which minimize measurement error (i.e. larger sample and use of multi-item indicators) should yield more representative networks, and have greater power to detect differences between networks, *should* differences exist at the population level.

Figure A1

Differences in Adjacency Matrices: Edges Estimated as Present versus Absent



Note. Effects of sample size and node aggregation on differences in adjacency matrices, across resampling conditions: sampling variability (left), scale variability (middle), sampling and scale variability (right). Results are displayed in the form of stacked bar charts, such that each bar height represents the total number of possible network edges (i.e., 15), and coloured bar segments break down the relative proportions of total edges that were consistently estimated as absent (red bars) or present (blue bars), or different (green bars).

Table A1

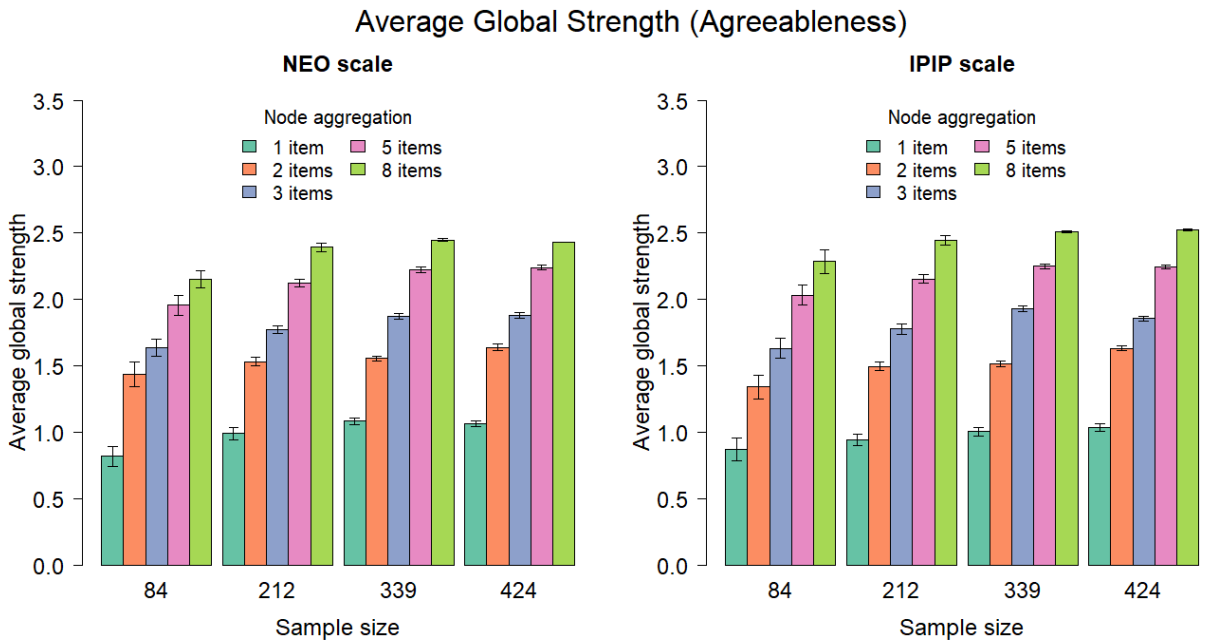
Differences in Adjacency Matrices: Descriptives [a]

	Sampling variability					
	Absent in both		**Present in both**		**Different in each**	
	***n* = 84**	***n* = 212**	***n* = 84**	***n* = 212**	***n* = 84**	***n* = 212**
1-item	9.42	8.34	1.46	2.96	4.12	3.70
2-item	7.80	7.00	2.10	4.30	5.10	3.70
3-item	7.18	5.80	2.86	5.20	4.96	4.00
5-item	6.22	4.94	3.84	6.00	4.94	4.06
8-item	5.46	3.54	4.04	6.62	5.50	4.84
	Scale variability					
	Absent in both		**Present in both**		**Different in each**	
	***n* = 84**	***n* = 212**	***n* = 84**	***n* = 212**	***n* = 84**	***n* = 212**
1-item	9.88	7.58	0.76	1.36	4.38	6.06
2-item	7.56	5.38	1.90	3.16	5.54	6.46
3-item	6.92	4.62	2.52	4.20	5.56	6.18
5-item	6.14	3.94	3.80	5.62	5.06	5.44
8-item	5.96	3.82	4.68	6.98	4.36	4.20
	Sampling and scale variability					
	Absent in both		**Present in both**		**Different in each**	
	***n* = 84**	***n* = 212**	***n* = 84**	***n* = 212**	***n* = 84**	***n* = 212**
1-item	9.70	7.40	0.78	1.58	4.52	6.02
2-item	7.60	5.66	1.78	3.14	5.62	6.20
3-item	6.52	5.04	2.23	4.32	6.16	5.64
5-item	5.90	3.98	3.38	5.68	5.72	5.34
8-item	5.68	3.94	4.12	6.86	5.20	4.20

a. Values correspond to Figure A1 above.

Figure A2

Average Global Strength



Note. Effects of sample size and node aggregation on average global strength of networks. Neuroticism networks were estimated via use of the NEO scale (left) and the IPIP scale (right). Error bars represent 95% confidence intervals around the mean.

Table A2

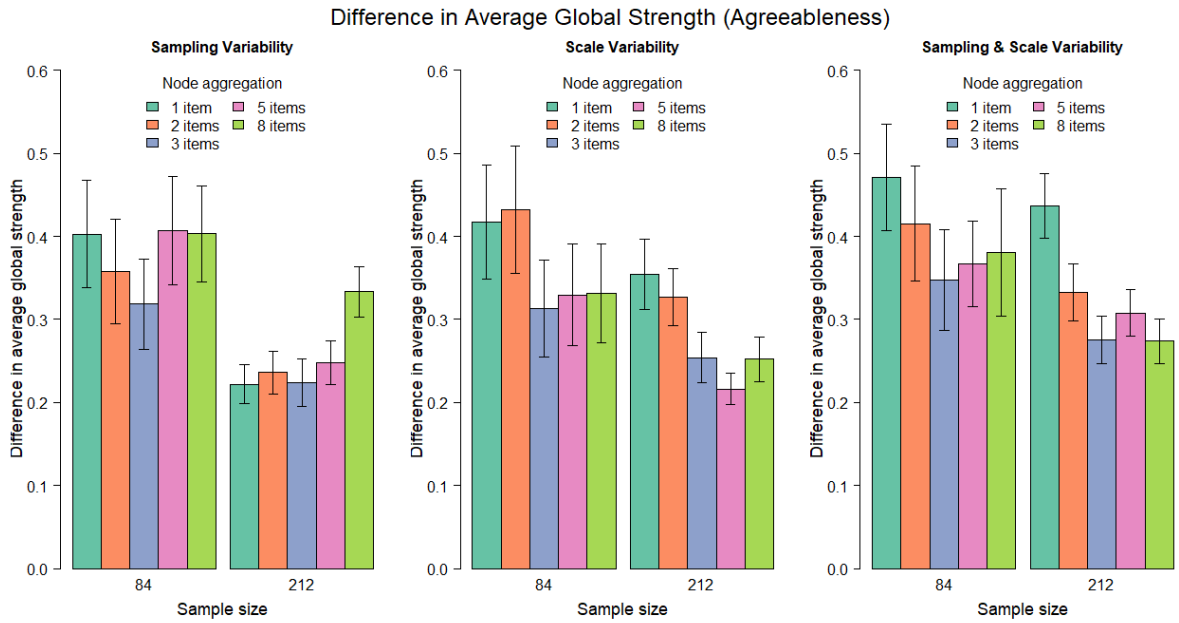
Average Global Strength: Descriptives [a]

	NEO scale				IPIP scale			
	***n* = 84**	***n* = 212**	***n* = 339**	***N* = 424**	***n* = 84**	***n* = 212**	***n* = 339**	***N* = 424**
1-item	0.82	0.98	1.08	1.06	0.87	0.94	1.00	1.03
2-item	1.43	1.53	1.55	1.64	1.34	1.49	1.51	1.63
3-item	1.64	1.77	1.87	1.88	1.63	1.78	1.93	1.86
5-item	1.96	2.12	2.23	2.24	2.03	2.15	2.25	2.24
8-item	2.15	2.39	2.45	2.43	2.28	2.44	2.51	2.52

a. Values correspond to Figure A2 above.

Figure A3

Differences in Average Global Strength



Note. Effects of sample size and node aggregation on differences in average global strength, across resampling conditions: sampling variability (left), scale variability (middle), sampling and scale variability (right). Error bars represent 95% confidence intervals around the mean.

Table A3

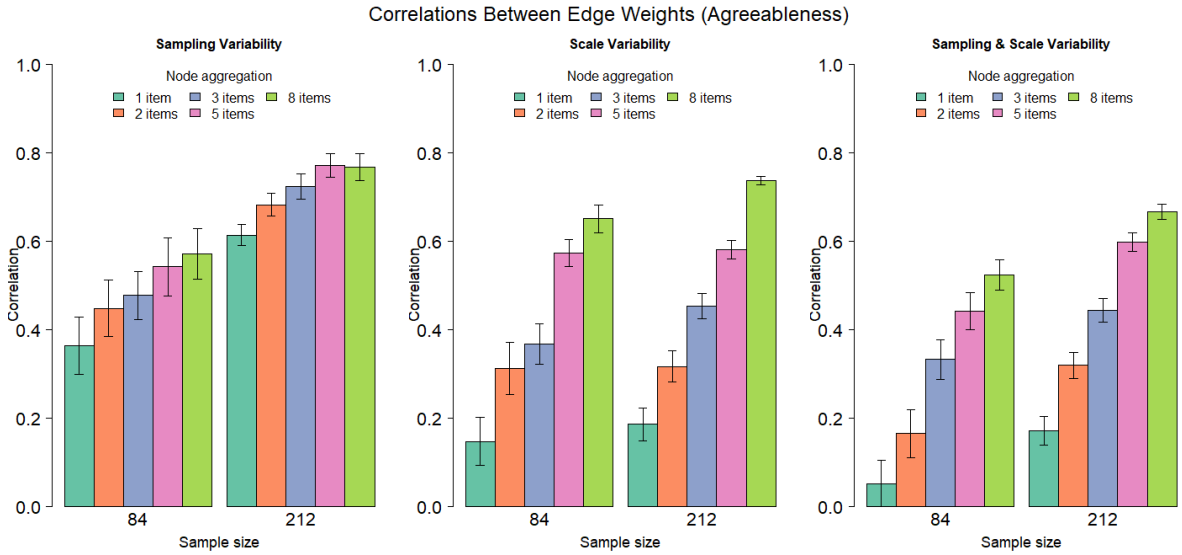
Differences in Average Global Strength: Descriptives [a]

	Sampling variability		**Scale variability**		**Sampling and scale variability**	
	***n* = 84**	***n* = 212**	***n* = 84**	***n* = 212**	***n* = 84**	***n* = 212**
1-item	0.40	0.22	0.42	0.35	0.47	0.43
2-item	0.36	0.24	0.43	0.33	0.42	0.33
3-item	0.32	0.22	0.31	0.25	0.35	0.28
5-item	0.41	0.25	0.33	0.22	0.37	0.31
8-item	0.40	0.33	0.33	0.25	0.38	0.27

a. Values correspond to Figure A3 above.

Figure A4

Correlations Between Edge Weights



Note. Effects of sample size and node aggregation on correlations between edge weights, across resampling conditions: sampling variability (left), scale variability (middle), sampling and scale variability (right). Error bars represent 95% confidence intervals around the mean.

Table A4

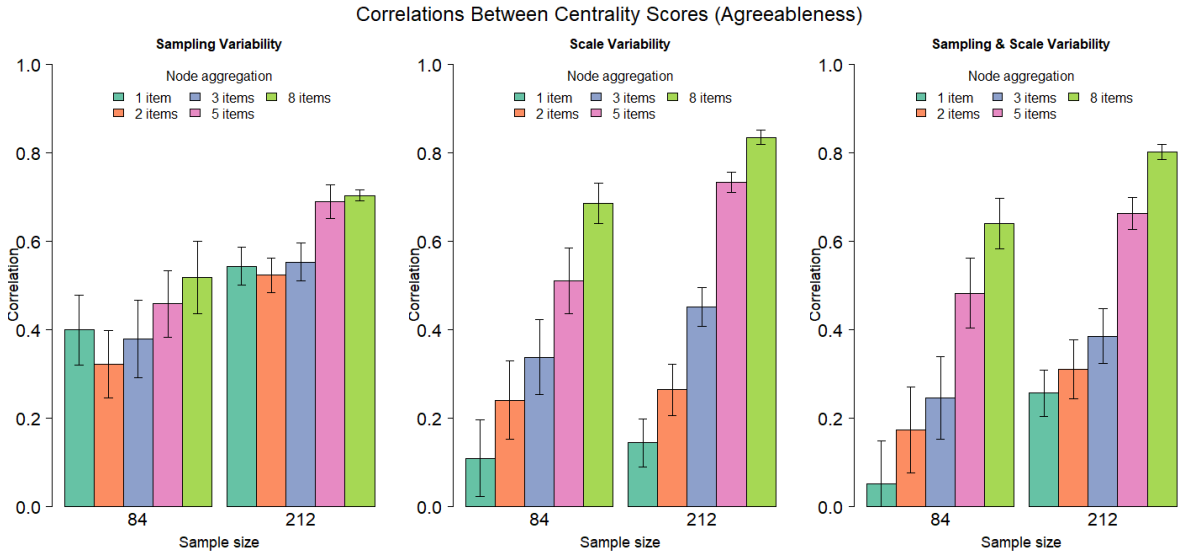
Correlations Between Edge Weights: Descriptives [a]

	Sampling variability		**Scale variability**		**Sampling and scale variability**	
	***n* = 84**	***n* = 212**	***n* = 84**	***n* = 212**	***n* = 84**	***n* = 212**
1-item	0.36	0.61	0.14	0.19	0.05	0.17
2-item	0.45	0.68	0.31	0.32	0.17	0.32
3-item	0.48	0.72	0.37	0.45	0.33	0.44
5-item	0.54	0.77	0.57	0.58	0.44	0.60
8-item	0.57	0.77	0.65	0.74	0.52	0.67

a. Values correspond to Figure A4 above.

Figure A5

Correlations Between Centrality Scores (Expected Influence)



Note. Effects of sample size and node aggregation on rank correlations between centrality (i.e. Expected Influence) scores, across resampling conditions: sampling variability (left), scale variability (middle), sampling and scale variability (right). Error bars represent 95% confidence intervals around the mean.

Table A5

Correlations Between Centrality Scores: Descriptives [a]

	Sampling variability		**Scale variability**		**Sampling and scale variability**	
	***n* = 84**	***n* = 212**	***n* = 84**	***n* = 212**	***n* = 84**	***n* = 212**
1-item	0.40	0.54	0.11	0.14	0.05	0.26
2-item	0.32	0.52	0.24	0.26	0.17	0.31
3-item	0.38	0.55	0.34	0.45	0.25	0.39
5-item	0.46	0.69	0.51	0.73	0.48	0.66
8-item	0.52	0.70	0.69	0.84	0.64	0.80

a. Values correspond to Figure A5 above.

Network Generalizability

Whole-Sample Networks

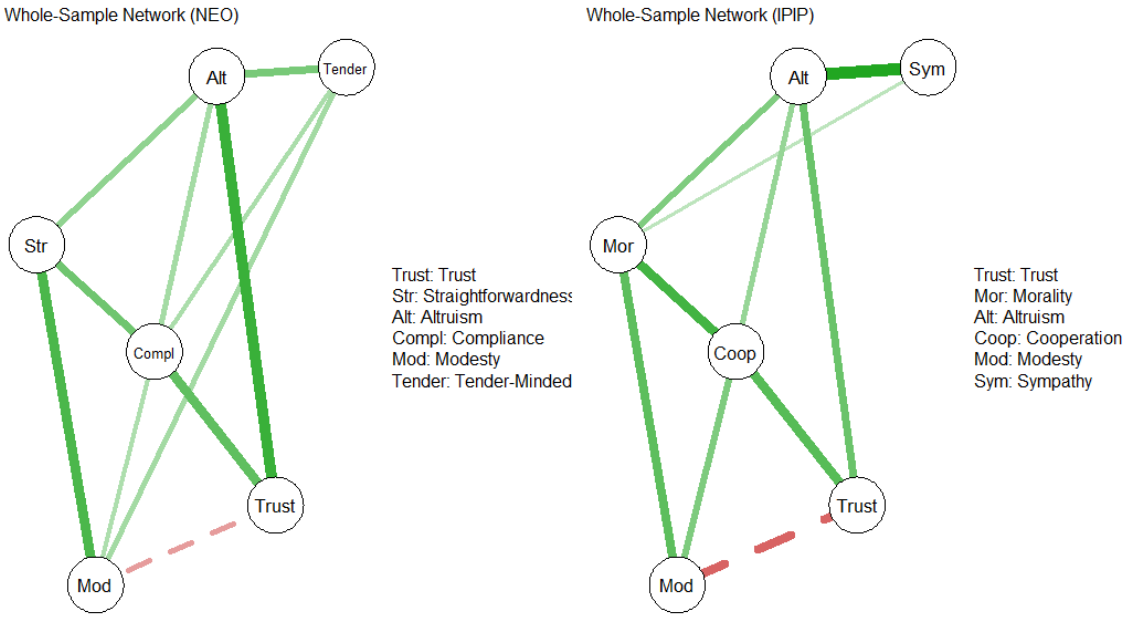
Whole-Sample Networks: Descriptive Statistics. Table A6 provides a descriptive summary of the network characteristics for each of the whole-sample networks (see Fig. A6).

Whole-Sample Networks: *NCT* Statistics. The *NCT* was run to test for the invariance of global strength, and the invariance of individual edges. The *NCT* failed to reject the test for invariance in global strength (difference = 0.17, p = .177). With respect to edge differences, the strongest edge was not consistent across both networks. In the NEO network, the edge linking the Trust and Altruism facets was strongest; this weight was weaker in the IPIP network (edge weights of 0.35 and 0.26, respectively). In the IPIP network, the strongest edge was between Altruism and Tender-Mindedness (or Sympathy); this weight was weaker in the NEO network (edge weights of 0.39 and 0.24, resp.). The largest difference in edge weights was the edge connecting Modesty and Tender-Mindedness (difference = 0.16; p = n.s.). Overall, across all 15 pairs of edges, the *NCT* failed to reject the hypothesis of invariance in edge strengths.

Whole-Sample Networks: Correlational Analyses. Correlation between edge weights was strong (r_p = .80), and correlation between centrality scores was very strong (r_s = .94); only the rank order of the two first nodes (i.e. Compliance and Altruism) did not correspond (see Fig. A8 and Table A6). *CS*-coefficients demonstrated moderate to high levels of centrality stability (see and Fig. A9 and Table A6).

Figure A6

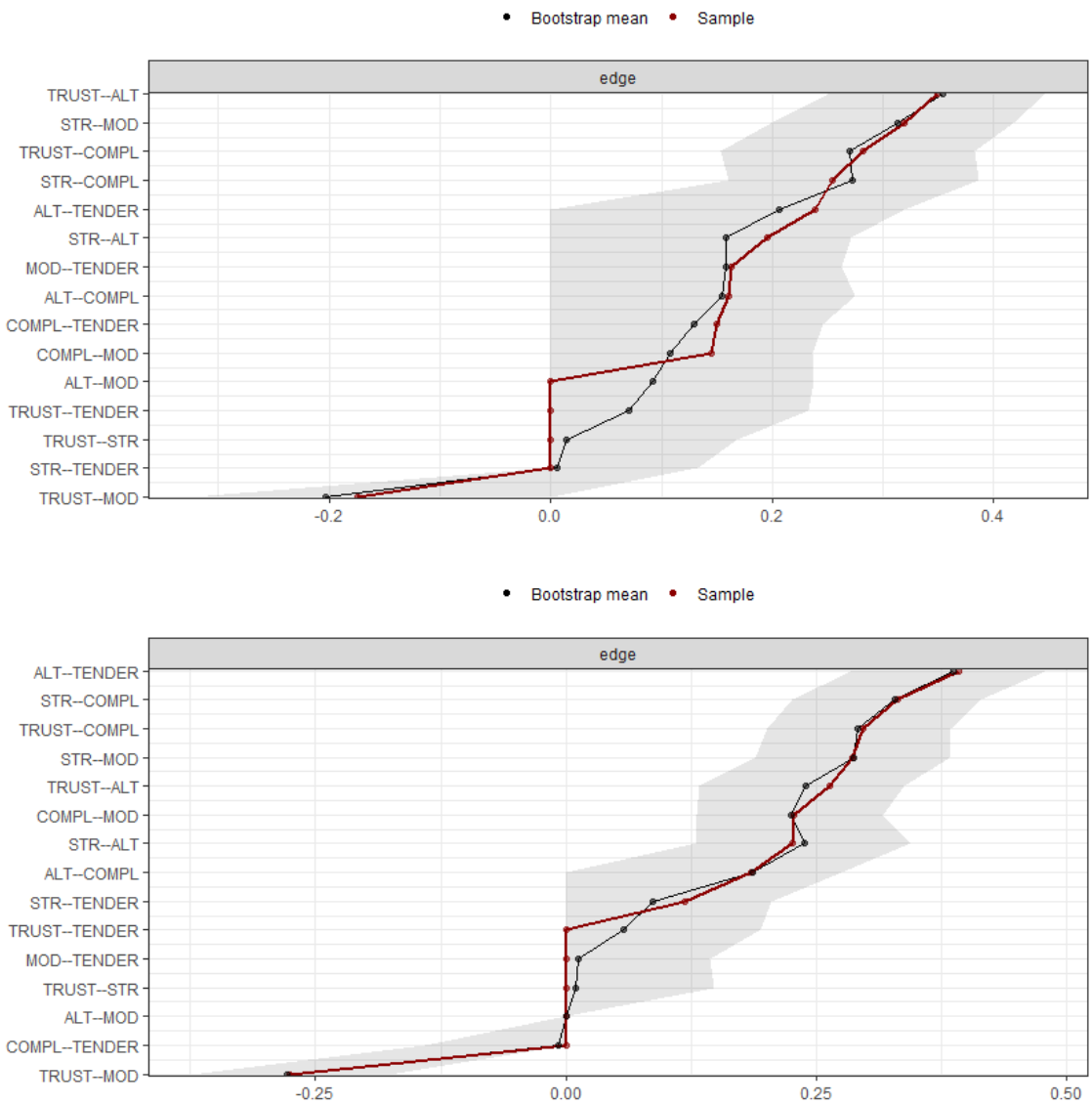
Whole-Sample Networks (Agreeableness)



Note. Non-regularized GGM networks for the Agreeableness network, estimated on the full sample (N = 424), using the NEO scale (left) vs. the IPIP scale (right). Solid-green lines represent positive edge weights; dashed-red lines represent negative edge weights. Networks were plotted using the average of the two Fruchterman-Reingold (or "spring") layouts in *qgraph* (max edge weight set to 0.45; Epskamp et al., 2012). Largest edge weights for the NEO and IPIP networks are, resp., between TRUST and ALT (edge weight = 0.35), and ALT and SYM (edge weight = 0.39).

Figure A7

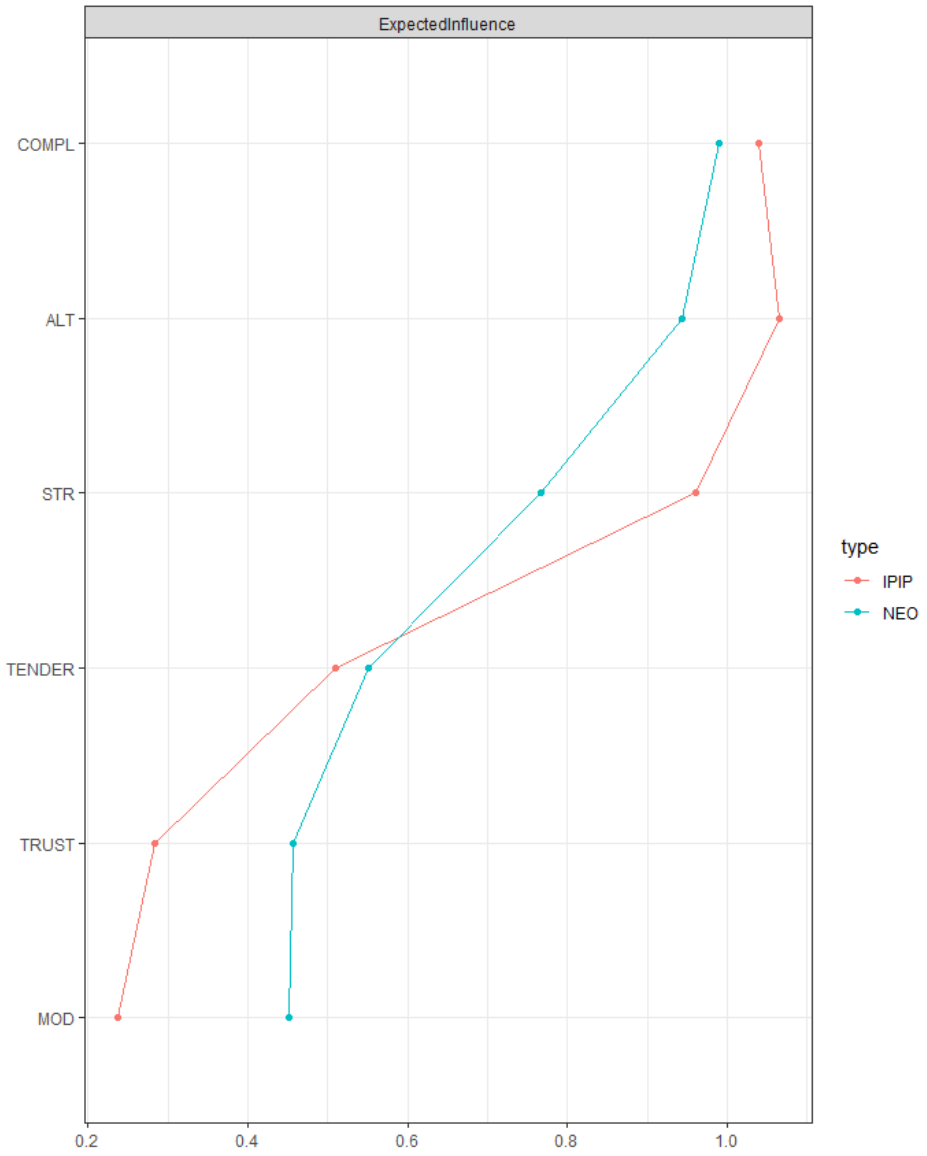
Whole-Sample Networks (Agreeableness): Accuracy of Edge Weight Estimates



Note. Edge weight estimates are plotted for NEO (top panel) and IPIP (bottom panel) whole-sample networks. All 15 edges are listed along the y-axis, in descending order of edge weights (indicated by the solid red line). Grey horizontal bars represent the 95% confidence intervals (CIs) for the estimates. Edges whose bootstrapped CIs contain zero should be interpreted with caution (see Epskamp et al., 2018). TRUST = Trust, STR = Straightforwardness, ALT = Altruism, COMPL = Compliance, MOD = Modesty, and TENDER = Tender-Mindedness.

Figure A8

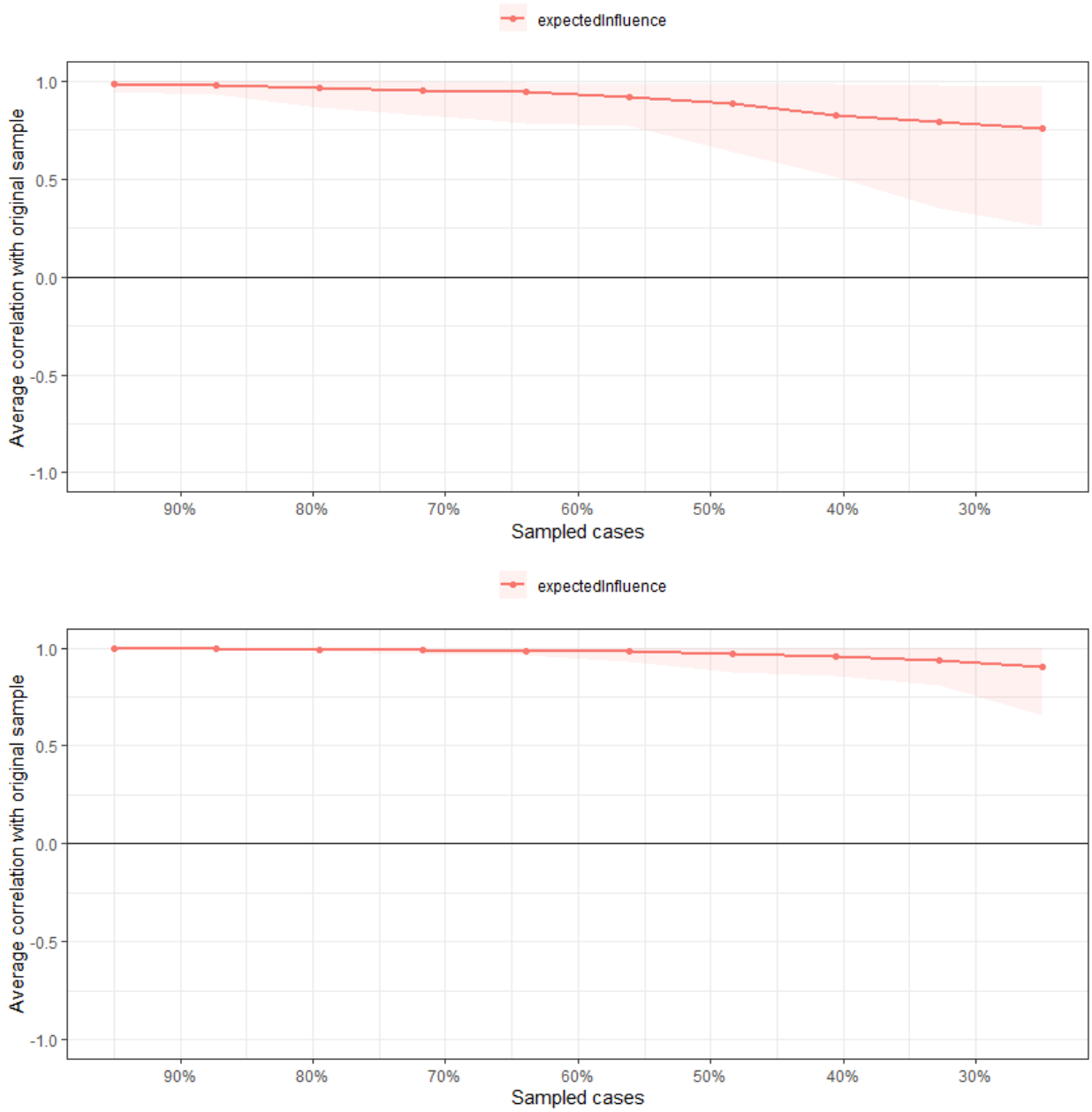
Whole-Sample Networks (Agreeableness): Correspondence in Expected Influence Scores



Note. Expected Influence scores for whole-sample networks are plotted in terms of raw values along the x-axis, with nodes ordered by (descending) rank along the y-axis. Blue line represents estimates derived from the NEO scale, and red line represents estimates derived from the IPIP scale. All values were computed on the same sample of subjects (i.e. full sample of N = 424). TRUST = Trust, STR = Straightforwardness, ALT = Altruism, COMPL = Compliance, MOD = Modesty, and TENDER = Tender-Mindedness. Most central node, as estimated by the NEO and IPIP scales, respectively, are Compliance and Altruism.

Figure A9

Whole-Sample Networks (Agreeableness): Centrality Stability (Expected Influence)



Note. Stability in Expected Influence scores are illustrated for the NEO-PI-R (top plot) and IPIP (bottom plot) population networks. X-axis represents proportions of sampled cases, and y-axis correlation coefficient values. Average correlations between centrality scores, estimated on the original sample and on the subsamples, are indicated by the solid line; shaded area indicates the range from the 2.5th to the 97.5th quantiles.

Table A6

Whole-Sample Networks: Descriptive Overview of Network Characteristics

		NEO	IPIP
Network characteristic		8-items	10-items
Global strength		2.43	2.60
Average edge strength: M (SD)		0.19 (0.14)	0.20 (0.18)
Number of non-zero edges		11	10
Number of zero edges		4	5
Network density		0.73	0.67
Largest edge weight		ALT–TRUST (0.35)	ALT–TENDER (0.39)
Strength of most different edge (MOD–TENDER)		0.16	0.00
CS-coefficient for Expected Influence [a]		0.52	0.75
Centrality ranks (Expected Influence raw score)			
	1st	COMPL (0.99)	ALT (1.06)
	2nd	ALT (0.94)	COMPL (1.04)
	3rd	STR (0.77)	STR (0.96)
	4th	TENDER (0.55)	TENDER (0.51)
	5th	TRUST (0.46)	TRUST (0.28)
	6th	MOD (0.45)	MOD (0.24)

Note. All values were computed on the full sample of N = 424. TRUST = Trust, STR = Straightforwardness, ALT = Altruism, COMPL = Compliance, MOD = Modesty, and TENDER = Tender-Mindedness.

a. *CS*-coefficients are an index of centrality stability, and represent the maximum proportion of sampled cases that can be dropped in order to retain a correlation of 0.7 (with 95% probability) between original sample and subsample centrality estimates (Epskamp et al., 2018).

Table A7

Scale Variability Condition: Descriptive Overview of Network Characteristics

Network characteristic Mean (SD)	1-item		3-items		8-items	
	NEO	IPIP	NEO	IPIP	NEO	IPIP
Average global strength	0.99 (0.33)	0.94 (0.32)	1.77 (0.21)	1.78 (0.27)	2.39 (0.24)	2.44 (0.24)
Average edge strength [a]	0.21 (0.04)	0.20 (0.05)	0.24 (0.03)	0.23 (0.04)	0.23 (0.04)	0.23 (0.04)
Mean no. of non-zero edges (out of a possible 15)	4.50 (1.47)	4.37 (1.32)	7.24 (1.22)	7.34 (1.42)	9.32 (1.36)	8.84 (0.17)
Mean no. of zero edges (out of a possible 15)	10.50 (1.47)	10.72 (1.44)	7.76 (1.22)	7.66 (1.42)	5.68 (1.36)	6.16 (1.17)
Avg. network density	0.3	0.29	0.48	0.49	0.62	0.59
Avg. correlation between edge weights	r_p = .19 (.27)		r_p = .45 (.21)		r_p = .74 (.07)	
Avg. correlation between centrality scores	r_s = .14 (.41)		r_s = .45 (.32)		r_s = .84 (.12)	
Centrality ranks: Expected Influence (frequencies, out of 50, listed in parentheses)						
Most central node (i.e. ranked 1st most often)	COMPL (13)	ALT (15)	ALT (20)	ALT (19)	ALT (28)	ALT (23)
Second most central node (i.e. ranked 1st second most often)	TRUST ALT (11)	COMPL (14)	COMPL (18)	STR (17)	COMPL (14)	COMPL (16)
Third most central node (i.e. ranked 1st third most often)	STR (8)	STR (11)	TRUST (5)	COMPL (12)	STR (8)	STR (11)
Largest edge weights (frequencies, out of 50, listed in parentheses)						
Largest edge	STR–MOD (11)	ALT–TENDER (12)	TRUST–ALT (19)	ALT–TENDER (13)	TRUST–ALT (22)	ALT–TENDER (28)

Second largest edge	STR–COMPL (9)	STR–COMPL (8)	TRUST–COMPL (8)	STR–COMPL (10)	STR–MOD (13)	STR–COMPL (15)
Third largest edge	TRUST–COMPL (8)	STR–ALT (5)	STR–COMPL (7)	ALT–COMPL (6)	STR–COMPL (8)	STR–MOD (3)
Maximum edge differences (frequencies, out of 50, listed in parentheses)						
Most different edge	ALT–TENDER (7)		ALT–TENDER TRUST–ALT (9)		COMPL–MOD (8)	
Second most different edge	TRUST–ALT STR–ALT STR–MOD (6)		TRUST–COMPL (7)		STR–ALT (7)	
Third most different edge	COMPL–TENDER (5)		STR–ALT (5)		ALT–TENDER TRUST–MOD TRUST–ALT (6)	

Note. All values were computed on a sample of n = 212. TRUST = Trust, STR = Straightforwardness, ALT = Altruism, COMPL = Compliance, MOD = Modesty, and TENDER = Tender-Mindedness. r_p and r_s denote Pearson and Spearman-rank correlation coefficients, respectively.

a. Average edge strength was computed as the mean of absolute edge weights across all non-zero edges in the network.

Appendix

NEO instrument (8 items per facet)	IPIP instrument (10 items per facet)
Trust (A1) (alpha = 0.84, omega = 0.84)	**Trust (A1) (alpha = 0.84, omega = 0.84)**
I believe that most people are basically well-intentioned.	Believe that others have good intentions.
I'm suspicious when someone does something nice for me. [R]	Suspect hidden motives in others. [R]
My first reaction is to trust people.	Trust others.
I have a good deal of faith in human nature.	Believe in human goodness.
I believe that most people will take advantage of you if you let them. [R]	Think that all will be well.
I think most of the people I deal with are honest and trustworthy.	Believe that people are basically moral.
I tend to be cynical and skeptical of others' intentions. [R]	Am wary of others. [R]
I tend to assume the best about people.	Trust what people say.
	Distrust people. [R]
	Believe that people are essentially evil. [R]
Straightforwardness (A2) (alpha = 0.76, omega = 0.78)	**Morality (A2) (alpha = 0.73, omega = 0.73)**
If necessary, I am willing to manipulate people to get what I want. [R]	Use others for my own ends. [R]
I'm not crafty or sly.	Use flattery to get ahead. [R]
Being perfectly honest is a bad way to do business. [R]	Know how to get around the rules. [R]
I would hate to be thought of as a hypocrite.	Obstruct others' plans. [R]
I couldn't deceive anyone even if I wanted to.	Put people under pressure. [R]
Sometimes I trick people into doing what I want. [R]	Pretend to be concerned for others. [R]
At times I bully or flatter people into doing what I want them to. [R]	Would never cheat on my taxes.
I pride myself on my shrewdness in handling people. [R]	Take advantage of others. [R]
	Cheat to get ahead. [R]
	Stick to the rules.

Altruism (A3) **(alpha = 0.73, omega = 0.74)**	**Altruism (A3)** **(alpha = 0.81, omega = 0.81)**
I go out of my way to help others if I can.	Love to help others.
I try to be courteous to everyone I meet.	Make people feel welcome.
I generally try to be thoughtful and considerable.	Am concerned about others.
Most people I know like me.	Look down on others. [R]
Some people think of me as cold and calculating. [R]	Turn my back on others. [R]
I'm not known for my generosity. [R]	Have a good word for everyone.
I think of myself as a charitable person.	Take no time for others. [R]
Some people think I'm selfish and egotistical. [R]	Anticipate the needs of others.
	Am indifferent to the feelings of others. [R]
	Male people feel uncomfortable. [R]

Compliance (A4) **(alpha = 0.73, omega = 0.73)**	**Cooperation (A4)** **(alpha = 0.71, omega = 0.72)**
I hesitate to express my anger even when it's justified.	Can't stand confrontations.
When I've been insulted, I just try to forgive and forget.	Hold a grudge. [R]
If someone starts a fight, I'm ready to fight back. [R]	Get back at others. [R]
I can be sarcastic and cunning when I need to be. [R]	Have a sharp tongue. [R]
I would rather cooperate with others than compete with them.	Insult people. [R]
If I don't like people, I let them know it. [R]	Love a good fight. [R]
I'm hard-headed and stubborn. [R]	Contradict others. [R]
I often get into arguments with my family and coworkers. [R]	Yell at people. [R]
	Am easy to satisfy.
	Hate to seem pushy.

Modesty (A5) **(alpha = 0.76, omega = 0.76)**	**Modesty (A5)** **(alpha = 0.77, omega = 0.77)**
I have a very high opinion of myself. [R]	Have a high opinion of myself. [R]
I'm better than most people, and I know it. [R]	Believe that I am better than others. [R]
I don't mind bragging about my talents and accomplishments. [R]	Boast about my virtues. [R]
I would rather praise others than be praised myself.	Dislike being the center of attention.
I try to be humble.	Consider myself an average person.

I'd rather not talk about myself and my achievements.	Dislike talking about myself.
I feel that I am no better than others, no matter what their condition.	Know the answers to many questions. [R]
I'm a superior person. [R]	Think highly of myself. [R]
	Seldom toot my own horn.
	Make myself the center of attention. [R]
Tender-Mindedness (A6) **(alpha = 0.61, omega = 0.61)**	**Sympathy (A6)** **(alpha = 0.76, omega = 0.77)**
I have sympathy for others less fortunate than me.	Feel sympathy for those who are worse off than myself.
I have no sympathy for panhandlers. [R]	Sympathize with the homeless.
I would rather be known as "merciful" than as "just".	Believe in an eye for an eye. [R]
We can do too much for the poor and elderly.	Try not to think about the needy. [R]
Political leaders need to be more aware of the human side of their policies.	Tend to dislike soft-hearted people. [R]
I'm hard-headed and tough-minded in my attitudes. [R]	Suffer from others' sorrows.
Human need should always take priority over economic considerations.	Believe that people should fend for themselves. [R]
I believe all human beings are worthy of respect.	Am not interested in other people's problems. [R]
	Value cooperation over competition.
	Can't stand weak people. [R]

Note. Reverse-items are denoted by [R].